

**Technological Institute of the
Philippines Computer Engineering
Department**

Enhancing Solid Waste Management in the Philippines: A Computer Vision Approach for Automated Garbage Segregation

In partial fulfilment of the requirements in
**Emerging Technologies 1 in CpE
(CPE018)**

Submitted To:
Engr. Jonathan V. Tylar

Group Members

**Castillo, Joshua
Cepe, Vince Paul
Cruz, Vince
Cuyugan, Emmanuel
De Leon, Juan Miguel Nathaniel**

Introduction

The project aims to create an object identifier program that detects whether the object shown in the camera is biodegradable or non-biodegradable. The program utilizes Keras Model as its deep learning algorithm, as well as OpenCV and NumPy. It is a practical solution that utilizes deep learning techniques that uses live feed from a camera.

Objective Of the Project

The program aims to help individuals that have a hard time identifying whether the object they have in possession belongs to which category, to properly dispose of it and help in implementing responsible waste disposal. The members also aim to leave their mark in reducing the garbage problem in the country by creating a program that can identify whether the object shown in the camera belongs to which bin.

Methodology

Our solution for waste segregation management is to use computer vision to recognize whether the garbage is biodegradable or nonbiodegradable.

Illustration:



As shown, Castillo used two samples, first is the bottle of coke, when he shows it to the camera it should display “di-nabubulok” which means a Non-biodegradable waste, and then on the 2nd example Castillo used okra(vegetable) when he shows it to camera it should say “nabubulok” which means biodegradable waste.

Data and Results

For the data and results, we gather the peak confidence score of each sample and average it to get their accuracy percentage.

Here are the results:

Biodegradable samples used from our demonstration:	Confidence score:
Okra	100%
Tomato	92%
Donut	100%
Rotten bell pepper	100%
Carrots	100%

Average accuracy: 98.4% for Biodegradable.

Non-Biodegradable samples used from our demonstration:	Confidence score:
Disposable spoon	100%
Pet bottle	100%
Plastic bag	100%
Noodles sachet	100%
Noodles wrapper	100%

Average accuracy: 100% for Non-biodegradable.

Conclusion and Learning

In this project, our group has learned a lot and several important takeaways and lessons have been learned. The development and use of the item identification programs for automated trash segregation represent a significant improvement in handling the solid waste management difficulties faced by the Philippines. Through the use of modern tools like Numpy and OpenCV for image processing, in addition to Keras Model for deep learning, the project successfully shows the usefulness of computer vision in environmental sustainability. The system's capacity to help people dispose of garbage properly is demonstrated by its ability to accurately identify waste as biodegradable or non-biodegradable through a live video feed. This creative method supports national efforts to promote sustainable practices and lessen environmental pollution in addition to helping with proper trash management.

Effective integrations of deep learning algorithms with computer vision tools such as OpenCV show how these technologies may be used for ordinary activities like garbage sorting, allowing for the application of complex AI models to tackle real-world problems. The use of technology in trash management highlights the relevance of innovative solutions in tackling environmental concerns, since automated garbage segregation systems may considerably eliminate human error and enhance the efficiency of waste disposal operations. A software has to be simple to use and intuitive in order for technology to be broadly embraced. This emphasizes how important it is for technology solutions to be widely available and beneficial to the public in order to maximize its impact.

Furthermore, the project highlights the fundamental versatility of AI-based trash segregation systems for tackling a variety of changing difficulties by showing how these systems may be scaled and tailored to varied settings and waste management demands. The success of this project also serves as further evidence of the value of teamwork in the creation of new technologies, as integrating knowledge from areas like computer science, environmental science, and public policy may result in more comprehensive and practical solutions. In conclusion, this project not only helps the Philippines' waste management practices but also offers a useful framework for upcoming projects that seek to use technology to promote environmental sustainability. This will encourage more study and advancement in the field of automated waste segregation and beyond.

Progress Report

WEEK 1

- The group started its initial brainstorming. We talked about numerous ideas. But the one we decided on to pursue is the problem of segregating garbage through computer vision. With this, the group knew and decided that it can be a big help to society, especially to the people of the Republic of the Philippines.

WEEK 2

- This week is where the week started assigning the members different jobs. In this week we decided to have members for Coding, Documentation, and the Dataset. The members assigned are as follows: **DE LEON** and **CASTILLO** for **CODING**, **CEPE** and **CUYUGAN** for **DOCUMENTATION**, then **CRUZ** for the **DATASET**. With this, it is ensured that everyone will be participating in this project.

WEEK 3

- Progress for coding has been great as the group has found a way to implement computer vision in garbage segregation. Within the week, the group focuses on perfecting the code for the project. Help was expected from the members working on the documentation and the dataset in coding.

WEEK 4

- Finalization of the codes as well as working on the documentation. This is the work done on the whole week 4.

WEEK 5

- Finalizations of all the needed requirements for the project. The CVs, Video Presentation, PowerPoint, Documentation, and more. The project is finished and done.

Codes

```
from keras.models import load_model
import cv2
import numpy as np

# Disable scientific notation for clarity
np.set_printoptions(suppress=True)

# Load the model
model = load_model("C:/Users/James/AppData/Local/Temp/Rar$DRa3200.32238/keras_model.h5", compile=False)

# Load the labels
class_names = open("C:/Users/James/AppData/Local/Temp/Rar$DRa3200.33615/Labels.txt", "r").readlines()

# CAMERA can be 0 or 1 based on the default camera of your computer
camera = cv2.VideoCapture(0)
```

- call the keras which is the training model for the technique that we use for our dataset.
- import cv2 for object detection.
- Loaded the model of our trained dataset using h5 fil. We used the directory where it is located to properly loaded the trained model
- Loaded the text file for the labels of our data set we used the directory where it is located to properly loaded the labels.

```
while True:
    # Grab the webcam's image
    ret, image = camera.read()

    # Increase brightness using gamma correction
    gamma = 0.4
    image = np.power(image / 255.0, gamma)
    image = (255 * image).astype(np.uint8)

    # Resize the raw image into (224-height, 224-width) pixels
    image = cv2.resize(image, (224, 224), interpolation=cv2.INTER_AREA)

    # Make the image a numpy array and reshape it to the model's input shape
    image = np.asarray(image, dtype=np.float32).reshape(1, 224, 224, 3)

    # Normalize the image array
    image = (image / 127.5) - 1

    # Predict the model
    prediction = model.predict(image)
    index = np.argmax(prediction)
    class_name = class_names[index]
    confidence_score = prediction[0][index]

    # Format class name and confidence score
    class_name_formatted = class_name[2:]
    confidence_score_formatted = np.round(confidence_score * 100)
```

This code section here is for:

- First the brightness of the camera, we change the gamma to 0.4
- Size of the camera, we used 224x224 for the size
- Prediction model
- Format of the labels and the confidence score

```
# Draw class and confidence score on the image
label_class = f"{class_name_formatted}"
label_confidence = f"{confidence_score_formatted}%"
cv2.putText(image[0], label_class, (10, 30), cv2.FONT_HERSHEY_SIMPLEX, 1, (0, 255, 0), 2)
cv2.putText(image[0], label_confidence, (10, 70), cv2.FONT_HERSHEY_SIMPLEX, 1, (0, 255, 0), 2)

# Show the image in a window
cv2.imshow("Webcam Image", image[0])

# Print prediction and confidence score
print("Class:", class_name_formatted)
print("Confidence Score:", confidence_score_formatted, "%")

# Listen to the keyboard for presses
```

- label_class is the created class for the label of what object is being detected
- Label_confidence is the created class for the confidence level of detecting from 0-100%
- Cv2.imshow command to show the camera on the monitor
- print class will show on the camera the word "class" and beside is the label whether the object is "nabubulok" or "di nabubulok".
- Print confidence score will show on the camera the text "confidence score" and beside it will show the confidence level in the form of percentage (function of the confidence level class).

Links

Video demonstration:

https://drive.google.com/file/d/1aW3NU2A7JtMokFtTdw8tBes4UgrwARbx/view?fbclid=IwZXh0bgNhZW0CMTAAAR0G5J3g3NW1AT6BDfCWq_civjG6D7VqqjBdNFsLVOnjb_CAqARvXWNJYtU_aem_AdUTYfC0cKJYiHANV7zjM-2gmgT2jwC48mwluQ6ku4lb6f1-1nV3J8tX33S16ewnp0TI-kcfIV5t5O75GriMNzvs

Dataset:

https://drive.google.com/drive/folders/1F36nkj4mIQcl_AyxEZwfwe2gaFo8ynNQ?usp=sharing

Google Drive:

https://drive.google.com/drive/folders/1AEoLUoadeEKIaxuINGNSVipHVHH7tPxD?usp=drive_link

Task Distribution

Castillo

Coding and Troubleshooting

Cepe

Documentation and Troubleshooting

Cruz

Documentation and Troubleshooting

Cuyugan

Coding and Troubleshooting

De Leon

Documentation and Troubleshooting

CV



32-33 Melbourne st.
Cainta, Rizal 1900
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SKILLS

Project Management

Knowledge in Linux Distribution
specifically in ubuntu and CentOS

LANGUAGES

English, Filipino

Castillo, Joshua L.

EDUCATION

SSS-VES Elementary School, Marikina — Elementary

June 2006 - April 2012

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Our Lady of Perpetual Succor College, Marikina — High-School

June 2013 - April 2018

Technological Institute of the Philippines, Quezon City — BS Computer Engineering

June 2018 - on going

PROJECTS

Waste segregation management — waste segregation management using Computer vision



Cepe, Vince Paul J.

EDUCATION

St. Matthew College, San Mateo — Elementary

June 2006 - April 2012

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St. Matthew College, San Mateo — High-School

June 2013 - April 2018

Technological Institute of the Philippines, Quezon City — BS Computer Engineering

June 2018 - on going

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SKILLS

Python,C++ Programing

LANGUAGES

English, Filipino

PROJECTS

Waste segregation management — waste segregation management using Computer vision



Cruz, Vince Joseph B.

EDUCATION

Good Samaritan Christian School, Quezon City — Elementary

2008 - 2015

Hope Christian Academy, Quezon City — High School

2016 - 2019

CIIT Senior High School, Quezon City — Senior High School

2019 - 2021

Technological Institute of the Philippines, Quezon City — BS Computer Engineering

2021 - on going

23 Mangga St. Quezon City

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qvjbcruz@tip.edu.ph

SKILLS

Project Management

2D/3D Unity Game
Development

Video Editing

Embedded Systems

Programming Languages
(Python, C++)

LANGUAGES

English, Filipino

PROJECTS

**Waste segregation
management — waste
segregation management using
Computer vision**



Cuyugan, Emmanuel Luis B.

EDUCATION

Aquinas School, San Juan City — Elementary

2009 - 2015

Immaculate Heart of Mary College, QC — Junior High-School

2015 - 2019

Far Eastern University HS, Manila — Senior High-School

2019 - 2021

Technological Institute of the Philippines, Quezon City — BS Computer Engineering

2021 - on going

3 Interior 17 Lakas St.,
Matandang Balara, Quezon City 1119

09683251772

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SKILLS

Knowledge in Linux
Distribution specifically in
ubuntu and CentOS

Texture and Material Creating
for 3D Modelling

Knowledge in Programming
Languages such as C++ and
Python

LANGUAGES

English, Filipino

PROJECTS

**Waste segregation
management — waste
segregation management using
Computer vision**



De Leon, Juan Miguel Nathaniel

EDUCATION

Holy Redeemer School, Dasmarinas Cavite — Elementary

June 2008 - April 2009

Roosevelt College, Cainta Rizal — Elementary

June 2009 - April 2010

Sta Cecilia Parochial School, San Mateo Rizal — Elementary

June 2010 - April 2012

**St. Matthew College, San Mateo Rizal — Junior Highschool
High-School**

June 2012 - April 2016

La Salle Green Hills, Ortigas — Junior Highschool

June 2017 – July 2017

FEU Roosevelt Cainta, Cainta Rizal— Junior-Senior Highschool

June 2018 - March 2021

**Technological Institute of the Philippines, Quezon City — BS
Computer Engineering**

August 2021 - on going

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SKILLS

Automation utilizing NodeMCU and Arduino
Hardware

LANGUAGES

English, Filipino

PROJECTS

Waste segregation management —
*waste segregation management
using Computer vision*

Sleep Position Tracker — Sleep
*position tracker utilizing NodeMCU
and MPU6050 paired with a mobile
app*

CV G2 CPE018 CPE31S1